

BY 517: Introduction to Proteomics

Course Code	BY 517
Course Name	Introduction to Proteomics
L-T-P-C	3-0-0-3
Prerequisites	IC 136 - Understanding Biotechnology & its Applications
Intended for	Core for Dual Degree Bio Engg. students; Elective for other B.Tech and M.Tech students. B. Tech. 3rd and 4th year, MS/MSc. /M.Tech., Ph.D.
Mutual Exclusion	None
Approval	10th Senate (Jun 2016)
Revised	33rd BoA (Jan 2020) and 24th Senate (Feb 2020) - syllabus revised
Discontinued	60th BoA (Jun 2025)

Course Contents

- **An introduction to proteomics:** Basics of protein structure and function, An overview of systems biology, Evolution from protein chemistry to proteomics. (6 hours)
- **Abundance-based proteomics:** Sample preparation and prefractionation steps, Gel- based proteomics - two-dimensional gel electrophoresis (2-DE), Two-dimensional fluorescence difference in-gel electrophoresis (DIGE), Principles of Blue Native-PAGE, Staining techniques, Fundamentals of liquid-chromatography (LC) based protein and peptide separation methods. (6 hours)
- **Central role of mass spectrometry:** Different types of mass spectrometers with respect to source ionization and design, Different kinds of mass analyzers, Different mode of data acquisition; Concepts of top-down vs bottom-up approaches and targeted vs untargeted approaches in proteomics. (6 hours)
- **Quantitative proteomics:** stable isotope labeling by amino acids in cell culture (SILAC), Isotope-coded affinity tag (ICAT), Isobaric tagging based methods for quantitative proteome analyses (iTRAQ/TMT), Label free quantitation (LFQ) (MS based, data-independent acquisition-DIA etc.), Targeted approaches (SRM,MRM). Challenges in performing proteomics in biofluids such as plasma, serum etc. (clinical proteomics). (6 hours)
- **Functional proteomics:** Recombinational cloning, Protein-protein interaction techniques by yeast two-hybrid, immunoprecipitation, protein microarrays, Nucleic Acid Programmable Protein Array (NAPPA), Surface Plasmon Resonance (SPR); Understanding post-translational modifications (PTMs) mainly phosphorylation and glycosylation. (6 hours)
- **Module VI:** Structural proteomics; Protein cross-link detection methods using mass spectrometry. (6 hours)
- **Module VII:** Bioinformatics in proteomics; Manual interpretation of typical mass spectra, Mass spectrometry big data analyses using open-source software suits; Challenges and future prospects of proteomics research. (6 hours)

Text Books

1. J. Lovric, **Introducing Proteomics: From Concepts to Sample Separation, Mass Spectrometry and Data Analysis**, John-Wiley & Sons, 2011.
2. David L. Nelson, Michael M. Cox., **Lehninger Principles of Biochemistry**, 7th Edition, Macmillan learning, 2017.

References

1. D.C. Liebler, **Introduction to Proteomics: Tools for the New Biology**, Humana Press, 2002.
2. T.D. Veenstra, J.R. Yates III, **Proteomics for Biological Discovery**, John-Wiley & Sons, 2006.
3. A.J. Link and J. LaBaer, **Proteomics: A Cold Spring Harbor Laboratory Course Manual**, Cold Spring Harbor Laboratory Press, 2009.
4. N.C. Mishra, **Introduction to Proteomics: Principles and Applications**, John-Wiley & Sons, 2010.
5. R. Matthiesen, **Mass Spectrometry Data Analysis in Proteomics**, Humana Press, 2020.
6. Recent research articles will be advised accordingly.